UNIVERSITY OF EASTERN PANGASINAN
COLLEGE OF INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Infrastructure Verification Checklist

Formal Quality Assurance, Audit & Compliance Matrix
Mission 10: Enterprise Cloud Infrastructure

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Student / Candidate:	**Carlo Nimer**
Year & Section:	Fourth Year – BSIT-4J
Course / Mission:	**CCM101 (Cloud Computing) — Final Requirement (Mission 10)**
Instructor / Evaluator:	Prof. Jenkielyn C. Torres
Academic Year:	2026–2027
Date of Submission:	October 18, 2026

Quality Assurance & Evaluation Guidelines

This verification checklist serves as the formal technical evaluation and audit rubric for the Enterprise Cloud Infrastructure project. It details the exact verification method, target requirements, expected criteria, and evidence artifacts required to confirm full compliance.

Instruction for Evaluator: The 'Actual Observed Result' and 'Pass / Fail' columns are intentionally provided blank to facilitate live hands-on auditing during the recorded technical defense. A requirement may only be marked as 'PASS' upon successful live execution and verification.

Comprehensive Infrastructure Audit Matrix

ID	Requirement Specification	Verification Method / Command	Expected Criteria	Actual Observed Result	P/F	Notes
CHK-01	Windows Host OS Audit	Get-ComputerInfo in PowerShell	Windows 11 64-bit platform confirmed			
CHK-02	Physical RAM Capacity	Get-CimInstance Win32_ComputerSystem	>= 16 GB Total Physical Memory			
CHK-03	Hypervisor Installation	VBoxManage --version in PowerShell	Oracle VM VirtualBox 7.x installed			
CHK-04	VM Definition & Hardware	VBoxManage showvminfo 'ubuntu-docker'	2 vCPUs, 4096 MB RAM, Bridged Network			
CHK-05	Out-of-Band SSH Access	ssh carlo@192.168.100.52 from PowerShell	Encrypted terminal session established			
CHK-06	Guest OS Release	hostnamectl on Ubuntu Server	Ubuntu Server 24.04 LTS (Noble Numbat)			
CHK-07	Guest Memory Allocation	free -h on Ubuntu Server	~3.8 GiB Usable RAM, 2.8 GiB+ Swap active			
CHK-08	Guest Virtual Storage	df -h / on Ubuntu Server	Root filesystem has > 50% free capacity			
CHK-09	UFW Perimeter Firewall	sudo ufw status verbose	Status: active; Allow 22/tcp & 8088/tcp			
CHK-10	Port 5000/5432	sudo ss -tulpn | grep -E	Zero host-bound listeners for			

	Isolation	'5000|5432'	5000/5432			
CHK-11	Docker Engine Runtime	docker version on Ubuntu Server	Docker Community Engine active			
CHK-12	Docker Compose Version	docker compose version	Docker Compose v2.x installed			
CHK-13	Compose Manifest Syntax	docker compose config -q	Exit code 0; Zero syntax/indentation errors			
CHK-14	Service Execution State	docker compose ps	All 3 services (nginx, app, db) 'Up'			
CHK-15	Container Healthchecks	docker inspect --format='{{.State.Health}}'	app and db report status: 'healthy'			
CHK-16	Nginx Reverse Proxy	curl -i http://127.0.0.1:8088/	HTTP/1.1 200 OK & web portal loaded			
CHK-17	Flask Application Logic	curl -i http://127.0.0.1:8088/health	HTTP 200 OK; JSON payload returned			
CHK-18	PostgreSQL Database Engine	docker compose exec -T db pg_isready	PostgreSQL accepting socket connections			
CHK-19	Transactional Insertion	Submit Workload form in Web Portal	New row committed to workloads table			
CHK-20	Named Volume Persistence	docker volume inspect postgres_data	Volume mounts to /var/lib/postgresql/data			
CHK-21	State Survival on Restart	docker compose restart db	Database reconnects with zero data loss			
CHK-22	Container Memory Limits	docker stats --no-stream	Memory usage strictly complies with limits			
CHK-23	Backup Script Permissions	ls -l automation-script.sh	Executable bit set (-rwxr-xr-x / 0755)			
CHK-24	Cron Job Configuration	crontab -l	Active trigger scheduled at 0 2 * * *			

CHK-25	Backup Archive Creation	ls -lah backups/cloudapp_*.sql.gz	Gzip archive generated (size > 2 KB)			
CHK-26	Archive Gzip Integrity	gzip -t backups/cloudapp_*.sql.gz	CRC-32 verification succeeds (exit code 0)			
CHK-27	Non-Destructive Restore	Restore dump into cloudapp_restore_test	Schema and table records restored 100%			
CHK-28	Documentation Integrity	Audit Word manuals and architecture diagram	Consistent ports, IPs, and topologies			

Formal Academic Evaluation & Sign-Off

This section formalizes the evaluation, technical defense review, and academic grading of the Enterprise Cloud Infrastructure project submission.

Candidate Student Name: Carlo Nimer

Candidate Section & Program: Fourth Year – BSIT-4J (BS Information Technology)

Candidate Signature & Date: _____ Date: _____

Instructor / Evaluator: Prof. Jenkielyn C. Torres

Final Verification Outcome: **[ ] PASSED / APPROVED** **[ ] REVISION REQUIRED**

Instructor Signature & Date: _____ Date: _____

Instructor Notes & Technical Defense Remarks: